

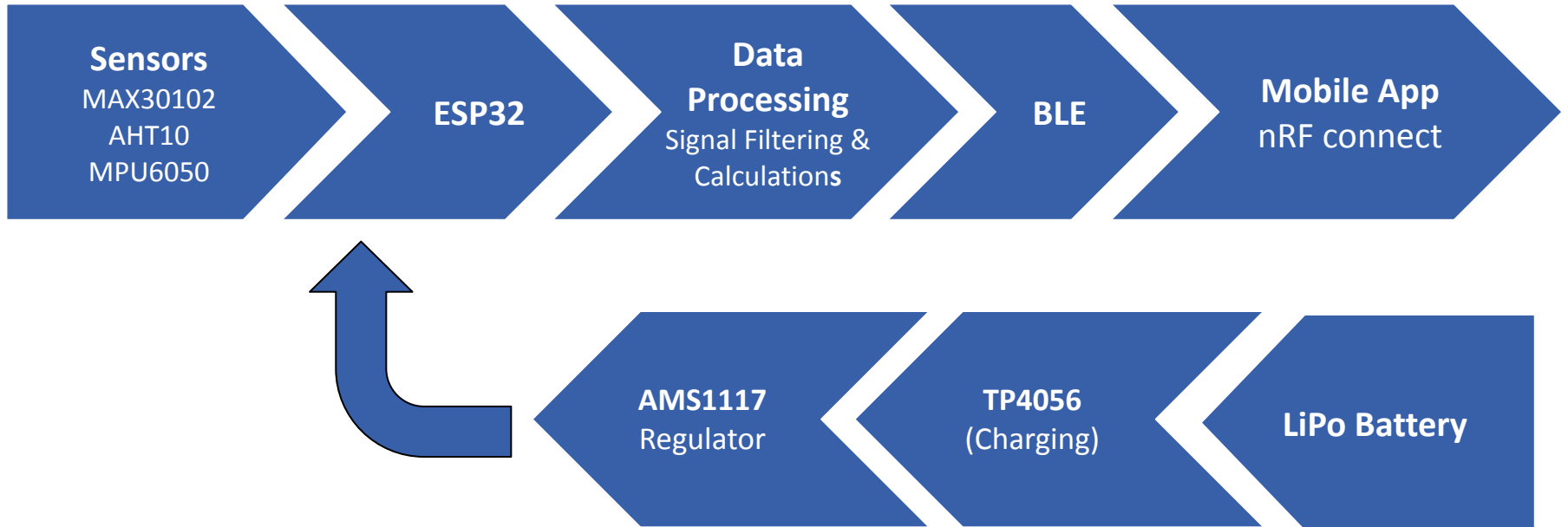
Sensor Integration and Communication Protocol

By Alle Keerthi Harsha

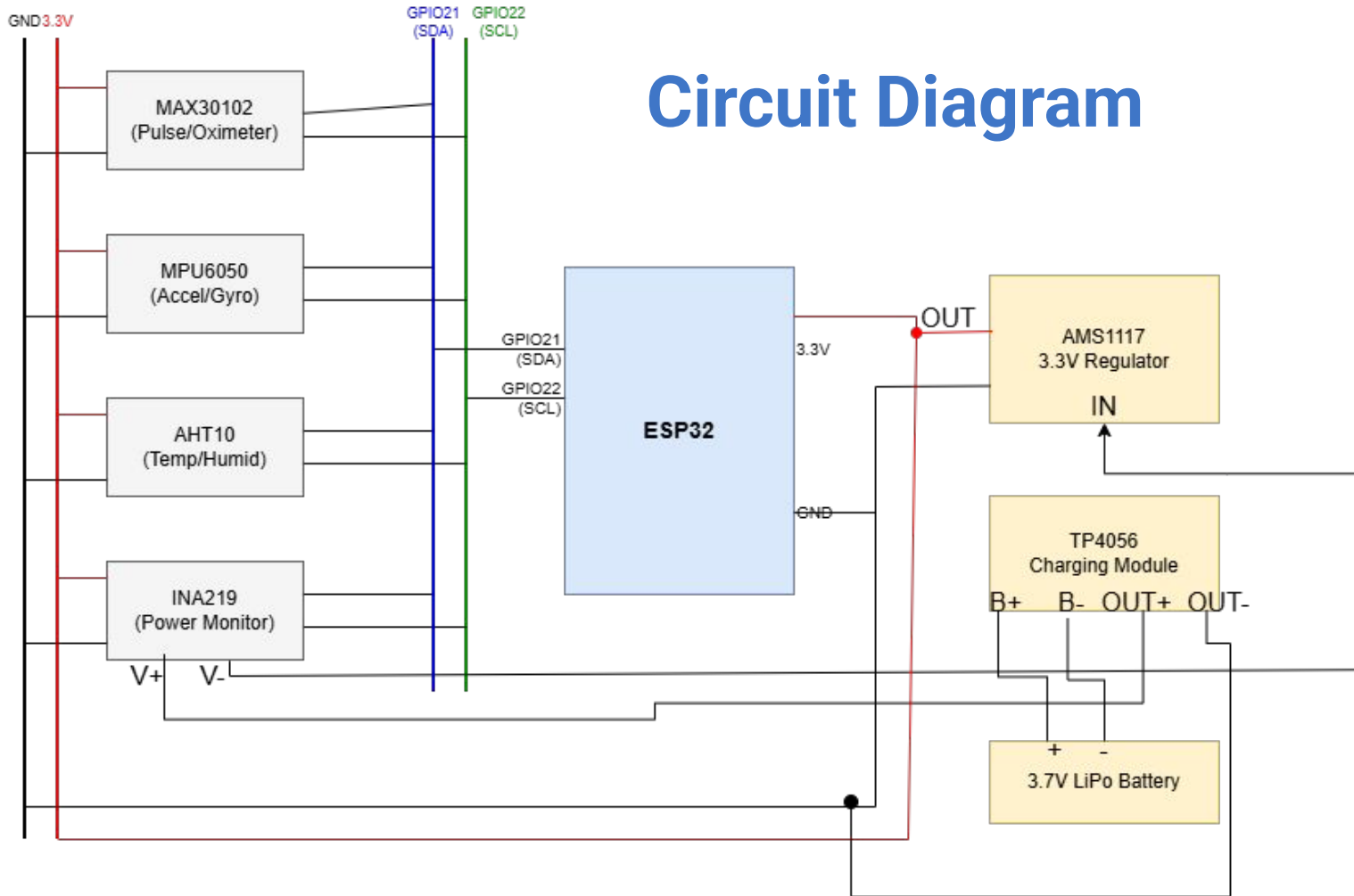
Features Planned

01	Real-time Heart Rate (BPM) monitoring	05	SpO ₂ (Blood Oxygen Level) measurement
02	Temperature monitoring	06	Battery monitoring and power management
03	Fall detection	07	Wireless data transmission via BLE
04	SOS / emergency alert system	08	Mobile application for live data display

System Overview



Circuit Diagram



Contribution

Sensor Integration

Interfacing MAX30102

Connected MAX30102 with ESP32 using I²C

Verified sensor detection (address check)

Initialized sensor and configured parameters

Data Analysis

Raw Signal Processing

Collected IR and Red signal data

Observed noise due to motion and placement

Identified instability in raw signals

Signal Processing

BPM & SpO₂ Calculation

Implemented peak detection for heart rate

Applied filtering (EMA, averaging)

Developed logic for stable BPM and SpO₂ output

Challenges

Signal Issues

Noisy Sensor Data

Raw IR and Red signals were highly unstable

Noise caused by finger movement and improper placement

Sudden spikes and drops affected accuracy

Processing Challenges

Filtering & Algorithm Tuning

Initial outputs were inconsistent without filtering

Required multiple changes in filtering logic (EMA, averaging)

Difficulty in selecting correct parameters for stable output

Implementation Issues

BPM Detection & Calibration

Incorrect Placement of finger

Needed proper tuning to avoid false detections

Incorrect values resulted in unrealistic BPM readings

Further Development

- Implementation of emergency alert (SOS) system
 - Integration with real-time notification services
 - Improved battery efficiency
 - AI-based health prediction
 - Miniaturization into wearable device
-

Thank you!
